

Target Trial Emulation of ARUBA

Emily Molins (DPhil) · for Truveta Data Engineering · July 2026

The management of an unruptured brain arteriovenous malformation (AVM), treat or monitor, is unresolved: the only randomized trial (ARUBA, 2014) favoured monitoring but is widely disputed, and a 2025 target trial emulation using a nationwide registry (MATCH) reached the opposite conclusion [1, 2]. We propose to re-examine this by emulating ARUBA in Truveta's US real-world data, adopting MATCH's operationalisation [2, 3]. This brief specifies the cohort and data required; the pivotal item is AVM severity, which lives in radiology reports rather than diagnosis codes.

Cohort

Include:

- **Adults**, age 18+ at the index date.
- **Incident, unruptured cerebral AVM** (ICD-10 Q28.2) with no AVM diagnosis and no intracranial haemorrhage (I60–I62) in a pre-index look-back window.
- **Observable at baseline** (continuous enrolment 6–12 months pre-index) to establish incidence and treatment-naïve status.

Exclude:

- **Ruptured presentation** (haemorrhage as the mode of, or preceding, AVM diagnosis).
- **Prior AVM treatment** (resection, embolisation, or radiosurgery) before the index date.
- **Insufficient baseline data** to define severity and covariates.

Each patient is assigned to interventional treatment initiated within 6 months of diagnosis, or conservative management; the type of intervention (microsurgery, embolisation, radiosurgery, or a combination) is tracked so outcomes can be analysed by modality. Primary outcome: death or symptomatic stroke, with hemorrhage-free survival at 5 and 10 years for comparison.

Table 1. Cohort, treatment, and outcomes (from codes)

Data element	Definition / values	Source in Truveta
Cohort entry	Unruptured brain AVM, incident diagnosis	ICD-10 Q28.2
No prior haemorrhage	No I60–I62 before index date	ICD-10 I60–I62 (look-back)
Treatment-naïve	No AVM procedure before index	Procedure codes (look-back)
Intervention arm	Surgery / embolisation / SRS within 6 months, by modality	CPT 61680–61692; 61624 (+S&I); 61796–61800; 77371–77373
Conservative arm	No intervention in grace period	Absence of procedure codes
Outcome	Death or symptomatic stroke; hemorrhage-free survival	Mortality linkage; ICD-10 I60–I63
Follow-up	Continuous enrolment per patient; target 5–10 y	Observation window; report median

Table 2. AVM severity (note-derived concepts, the critical ask)

Severity (the Spetzler–Martin grade) is read from imaging and lives in radiology reports, so it must be supplied as extracted concepts. The first three rows are the minimum needed to control confounding.

Data element	Definition / values	Source in Truveta
Nidus size	<3 / 3–6 / >6 cm	Radiology-report concept
Eloquent location	Eloquent vs non-eloquent	Concept
Deep venous drainage	Present / absent	Concept
Spetzler–Martin grade	Derived I–V (from the three above)	Concept if present; else derive

Data element	Definition / values	Source in Truveta
Nidus morphology	Diffuse vs compact	Concept
AVM location	Lobar / deep / infratentorial / brainstem	Concept
Associated aneurysm	Present / absent	Concept / ICD

Covariates: age, sex, presentation (seizure [G40], headache, incidental), comorbidities, calendar year, treating site.

What we would like Truveta to confirm

- Which severity components (size, eloquence, deep venous drainage, SM grade) exist as concepts, and at what coverage.
- Whether intervention type and AVM obliteration are captured.
- Cohort counts at each eligibility step.
- Available continuous follow-up, and whether all-cause mortality linkage is included.
- Cost and lead time to provision the Table 2 concepts.

References

1. Mohr JP, et al. ARUBA: a multicentre, non-blinded, randomised trial. Lancet 2014. (NCT00389181) <https://pmc.ncbi.nlm.nih.gov/articles/PMC4119885/>
2. Han H, et al. Interventional Treatment vs Conservative Management of Unruptured Brain AVMs (MATCH). JAMA Netw Open 2025;8(11):e2543408. <https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2841324>
3. Hernán MA, Robins JM. Using Big Data to Emulate a Target Trial. Am J Epidemiol 2016;183(8):758–764. <https://academic.oup.com/aje/article/183/8/758/1739860>